

A Revolution in Search of an Estimand: On the Idionomic Proposal of Hayes et al. (2026)

Magnus Johansson^{1*}, Matti Vuorre², Nils Hentati Isacsson³ & Kristoffer Magnusson¹

¹Centre for Psychiatry Research, Department of Clinical Neuroscience, Karolinska Institutet, Stockholm, Sweden

²Department of Social Psychology, Tilburg University, Netherlands

³Department of Medicine Huddinge, Karolinska Institutet, Stockholm, Sweden

2026-09-28

Hayes et al. (2026) argue that clinical psychology is stalled by an “ergodic error”, the application of group-level statistics to individuals, and propose idionomic analysis as the remedy. We agree that pooled associations do not need to describe individuals and that treatment effects vary between people, but the proposed remedy does not answer the clinical question. Personalizing treatment is a causal problem, yet Hayes et al. never state which causal quantity, or estimand, they target. Their evidence consists of observational within-person associations, which cannot separate effects from confounding that varies over time. Applied to observational data, none of the recommended methods can identify a causal effect. Non-ergodicity shows only that pooled and individual associations may differ, not that causal effects do. The reported heterogeneity is also questionable, in part because person-level estimates from short series are imprecise. The reliability Hayes et al. cite against this concern describes person means rather than within-person variation, and the statistics summarizing heterogeneity have not been adequately checked against simulated data in which there is no heterogeneity. Designs that randomize treatment within or between persons can target person-level and conditional effects, within limits set by carryover and irreversible change. Rather than the proposed idionomic revolution, clinical psychology needs to join the causal revolution, in which the estimand is stated before the data are analyzed.

Keywords: Causal inference, Estimands, Idiographic analysis, Treatment effect heterogeneity, N-of-1 trials

Hayes et al. (2026) argue that clinical psychology has stalled because it applies “ensemble-level statistics” (p. 1) to particular people without testing the conditions that would justify the inference, and they propose idionomic analysis as the remedy: modeling each person’s processes first, then generalizing from the person-level effects. They make several important observations. Pooled associations can fail to describe any given individual, and although causal effects typically vary between people when appropriately examined, their prediction intervals are rarely reported (Bolger et al., 2019; Borenstein et al., 2017).

Personalizing treatment requires two things: a causal quantity that answers the clinical question, and an estimate of it precise enough to guide the choice of treatment. The idionomic program as proposed provides neither. Hayes et al.’s central empirical claim rests on heterogeneity in observational within-person *associations*, while the conclusion concerns heterogeneity in *causal*

*Send correspondence to: Magnus Johansson, magnus.johansson.3@ki.se.

effects, and nothing they present supports the latter. Their second, descriptive claim, that individuals' associations differ as much as reported, is also unsupported. The person-level estimates are imprecise, and the heterogeneity statistics calculated from them have not been adequately checked against data without heterogeneity.

Though we disagree with Hayes et al.'s analysis, their vignette is informative. A clinician asks a client to give three ratings a day on four personally relevant items for 14 consecutive days, and finds that for this client experiential avoidance predicts sadness while valued action does not predict improved mood, and adjusts the treatment plan accordingly (Hayes et al., 2026, p. 10). Three questions decide whether that adjustment is warranted: what quantity the clinician needs, whether 42 occasions (three a day for 14 days) of observational data identify it, and whether they estimate it precisely enough to improve on the average. The second is a question of identification: can a causal quantity be determined from these data?

1 The estimand is never defined

The estimand is the target quantity of interest, and it should be stated before an estimator, such as a statistical model, is chosen (Kahan & Devane, 2026; Lundberg et al., 2021). Hayes et al.'s paper is about personalizing intervention, which is a claim about individual treatment effects. Four distinct target quantities appear without being distinguished. We write $Y_i(1)$ and $Y_i(0)$ for the outcomes person i would have with and without treatment, Y_{it} for the same on occasion t , and $\mathbb{E}$ for an average over persons, or over occasions when subscripted with t :

1. The average treatment effect (ATE), $\mathbb{E}[Y_i(1) - Y_i(0)]$, identified by randomizing persons in an RCT and pooling the individual results.
2. The conditional average treatment effect (CATE), $\mathbb{E}[Y_i(1) - Y_i(0) \mid X_i = x]$, for persons who share the characteristics x , such as baseline severity or age.
3. The individual average treatment effect (iATE), $\mathbb{E}_t[Y_{it}(1) - Y_{it}(0)]$, averaged over one person's occasions.
4. The within-person association β_i estimated by i-ARIMAX, which is not a causal estimand.

Hayes et al. demonstrate dispersion in β_i and conclude that the ATE is unfit for decisions about individuals, without saying whether the CATE or the iATE should replace it. A paper arguing that the field targets the wrong quantity is the last place to leave the target undefined. The vignette makes the gap concrete. The clinician needs the iATE, the effect of reducing avoidance on this client's sadness. What the vignette's data deliver is β_i , from 42 observational ratings.

In the vignette, $Y_{it}(1)$ and $Y_{it}(0)$ are the client's sadness on occasion t under an avoidance-targeted intervention and under a comparison condition. On any one occasion only one of the two can be observed, which is the fundamental problem of causal inference (Holland, 1986). The average across occasions is nonetheless identified when the two conditions are randomized across occasions and their effects do not carry over from one occasion to the next, a condition we return to below.

Where an intervention's effect persists after it ends, the iATE is identified for the person only under an assumption about how their untreated course would have continued, which their own data cannot check, and the target that randomization can identify is the CATE: the expected effect among patients who share this client's characteristics. Predictive approaches to heterogeneous treatment effects target this quantity (Kent et al., 2018; Kent et al., 2020), and psychotherapy

research has its own examples, such as the Personalized Advantage Index, which estimates it from treatment-by-covariate interactions in randomized trials (DeRubeis et al., 2014). Hayes et al. acknowledge that such matching improves outcomes but argue that it hits a ceiling, because the average gains of personalized care are small and prediction rules often fail to replicate (p. 13). Whether idiomically guided treatment does better is itself a causal question about groups, answerable by randomizing patients to idiomically guided or standard treatment.

2 The proposed methods have no identification strategy

Hayes et al. describe ARIMAX as estimating contemporaneous within-person links while “controlling trends and autocorrelation” (p. 11). Modeling trends, differencing, and moving-average structure corrects the *error structure* of the regression, and differencing removes confounding by shared trends. It does nothing about confounding that changes from occasion to occasion, which is the kind at issue here. Using a person as their own control can remove time-invariant confounders (Rohrer & Murayama, 2023), but it does not address confounders that vary from day to day within the person, such as sleep, stress, interpersonal conflict, medication changes, and life events.

Using ARIMAX, Hayes et al. show that associations differ across people not only in magnitude but in direction. Opposite directions across people is compatible with causal effect heterogeneity, but it is equally compatible with confounding that differs in size or direction from person to person, and ARIMAX cannot distinguish them. Hayes et al. do not discuss this alternative. Suppose valued action reduces sadness equally for everyone. Person A’s valued-action ratings are highest on the days they spend with a seriously ill parent, because being there is what they value, and those are their saddest days. That confounding is positive and larger than the effect, so their observed coefficient is positive. Person B does not experience such events, so theirs is negative. The coefficients have opposite signs but the causal effect is identical.

The other methods in Table 1 of Hayes et al. do not change this. Boruta and its variants rank predictors by random-forest variable importance (Kursa & Rudnicki, 2010), which carries no causal interpretation (Zhao & Hastie, 2021) and is unstable when predictors are correlated (Strobl et al., 2007). GIMME and S-GIMME select paths by stepwise search on modification indices (Gates et al., 2017; Gates & Molenaar, 2012) and then report uncertainty as though the model had been specified in advance (Berk et al., 2013). mIVAR is a descriptive network model (Bringmann et al., 2013) whose parameters do not support causal hypotheses without further assumptions (Ryan et al., 2022). GSOM has no outcome variable (Alahakoon et al., 2000), PECAN and PLAN elicit perceived rather than demonstrated causal structure (Klntwall et al., 2023), and idiomonic RE-MA pools the i-ARIMAX coefficients and inherits whatever they identify (Sahdra et al., 2024). In the tsBoruta tutorial, Sahdra et al. (2026) state that their findings describe how processes vary naturally rather than how they respond to manipulation, and call for micro-randomized or single-case experimental designs to verify that changing a target changes the outcome. They nonetheless define a process of change as a statistically significant predictor and report that targeting three processes could benefit 52 of 54 participants. Hayes et al. include the method in their proposal without the caveat stated by Sahdra et al. (2026).

The vocabulary in Hayes et al. includes terminology like “process of change,” “process-to-outcome link,” and “determinant.” These carry causal content while the formal claims remain associational, which leaves the causal inference implicit and its assumptions unexamined, to the detriment of scientific advance (Grosz et al., 2020). Assumptions that are never stated cannot

be examined, and the clinical recommendations are stated as though the causal effect had been established.

3 Non-ergodicity is not effect heterogeneity

Hayes et al. attribute the stall in clinical psychology to an “ergodic error.” A collection of persons is ergodic when each person’s observed process is stationary and identical across people (Molenaar, 2004). When these conditions hold, statistics computed across people at one occasion recover what following one person over time would give. Where the conditions fail, pooled associations need not describe individual associations (Fisher et al., 2018). This is correct, but it establishes only that they may fail to, not that they do or by how much. Moreover, both kinds of statistic are averages of an observed process, and neither says what would happen to a person under intervention. Heterogeneity in causal effects, the quantity the clinical argument requires, is not established by non-ergodicity.

Nor is the ergodic error needed to explain the stall. Mediation, which Hayes et al. list among the areas it has damaged, has a simpler diagnosis. Under the standard design, the indirect effect is identified only if the treatment and the mediator are each unconfounded given the same covariates, an assumption that does not follow from treatment randomization alone (Imai et al., 2010; Imai et al., 2011). Almost 20 years ago, Dunn & Bentall (2007) warned that causal assumptions are rarely stated, leaving thousands of mediation studies “of unknown and questionable value” (p. 4743). Lemmens et al. (2016) reviewed 35 studies of mediators in psychotherapy for depression against the criteria of Kazdin (2007). None met all six, and none manipulated the mediator experimentally. Their conclusion, that the central difficulty is “demonstrating the causal relation between change in the mediator and change in depressive symptoms,” names an identification failure, and none of it turns on ergodicity. Hayes et al. nonetheless propose ensemble-level mediational findings as Bayesian priors for person-level analysis, where they would pass their confounding on.

4 How much heterogeneity is there to explain?

Setting aside the causal issues, the paper makes a descriptive claim: that associations between processes and outcomes differ across individuals by a specific degree. Their Figure 2 is the central example, showing person-level associations between worrying that positive emotions are fading and happiness in 151 participants. If this magnitude is of importance, it is critical that it is correctly estimated and reported. However, the reported heterogeneity of person-level estimates combines real differences with estimation error, and the error is large at the series lengths the paper recommends. It is therefore unlikely that the degree of heterogeneity is accurately described by the proposed methods. The reliability evidence offered against this concern measures something else.

Hayes et al. write that “single- or few-item process and outcome measures have demonstrated excellent within-person reliability (ICC(2) commonly ≥ 0.90)” (p. 11), citing several papers. One reference is Sahdra et al. (2024), who report ICC(2) values of .92 to .95. But it is the reliability of the person means, not of within-person variation (LeBreton & Senter, 2008). ICC(2) is the between-person quantity ICC(1) scaled to the reliability of the mean by averaging over the repeated measurements, in this case around 20 to 25 per person, which is why it is high even when single-occasion reliability is modest (Cranford et al., 2006). The ICC(1) reported by Sahdra et al. (2024) ranges from .34 to .44 and implies that .56 to .66 of the variation lies within persons. That variance mixes true momentary change with measurement error and is the variation the idiographic models actually analyze (Cranford et al., 2006). A between-person mean reliability has been erroneously

re-labeled by Hayes et al. as within-person reliability. Comparable person-level reliability indices range from .27 to .35 when using 15 measurements (Rohrer et al., 2024). Advice built on estimates with such high levels of noise need not beat advice built on the average (Rohrer et al., 2024).

Hayes et al.'s τ is the between-person standard deviation of the underlying person-level coefficients. It is estimated by subtracting the average within-person sampling variance from the observed variance of person-level estimates and taking the square root. But the estimate is only as good as the standard errors fed into it, and those come from `auto.arima`, which selects a model and then reports uncertainty as though it had been fixed in advance (Berk et al., 2013). Standard errors that are too small leave too little to subtract, so $\hat{\tau}$, I^2 , and the prediction interval are all inflated. The same standard errors underlie Figure 2 of Hayes et al., where 58 of 151 participants show significant person-level effects, 38 of them opposite in direction to the average estimate. If everyone shared the average effect, at most about four significant results in the opposite direction would be expected at $\alpha = .05$. With correct standard errors, the 38 participants with opposite direction could not be due to sampling error, but whether the standard errors are correct was not evaluated in that study. Running the analysis with data simulated without heterogeneity and with realistic autocorrelation, series lengths, and item bounds would show how much of the count arises without any real heterogeneity. Sanford et al. (2022) ran such a check for the multilevel model on a single simulated dataset, whereas estimating how often a method reports heterogeneity that is not there requires hundreds of replicated datasets (Morris et al., 2019). The meta-analytic second step of i-ARIMAX is never run on those data. They run it only on empirical data, where the true heterogeneity is unknown, so their $\hat{\tau}$ and I^2 are never compared with a known value.

5 An idionomic program with an identification strategy

The problem Hayes et al. pose, estimating within-person effects from intensive longitudinal data when confounders vary over time, has an extensive literature that the paper does not engage with. N-of-1 randomized trials, which Hayes et al. mention approvingly and then set aside in favor of observational modeling, identify the iATE by randomizing the person's own treatment (Lillie et al., 2011). Micro-randomized trials randomize intervention delivery at each decision point and identify proximal effects averaged over people, and how those effects vary with time-varying context (Klasnja et al., 2015). Where randomization is impossible, g-methods and marginal structural models target contrasts between treatment histories, under assumptions that must be named, defended, and probed for fragility (Hernán & Robins, 2020; Robins et al., 2000). Where treatment is randomized within persons, effect heterogeneity is estimable and often found (Bolger et al., 2019; Haaf & Rouder, 2019; Vuorre et al., 2024).

In the vignette, the clinician could randomize days on which the client practices an avoidance-targeted skill against days on which they practice a comparison skill. The contrast in sadness would estimate the iATE for this client, with confounders such as sleep balanced by design rather than modeled. This is the alternating-treatments design that Hayes et al. themselves cite (Barlow & Hayes, 1979), and it works only if the skill's effect fades before the next assignment period.

That condition excludes much of psychotherapy. Exposure, cognitive restructuring, and new relational patterns work through learning that is cumulative, delayed, and cannot be withdrawn. Randomizing when a component is introduced, as in randomized multiple-baseline designs (Kratochwill & Levin, 2010), identifies the effect of introducing it, with the untreated course extrapolated from baseline. Sequential multiple-assignment randomized trials (Murphy, 2005) identify good adaptive treatment rules for people with similar characteristics as this client, a version of the

CATE, rather than effects for this particular client. For a treatment that cannot be withdrawn, no design shows what would have happened to this client without it. That is the fundamental problem of causal inference, not a shortcoming of any method, and it constrains the idionomic program as much as the designs proposed here.

6 Recommendations

1. State the estimand before choosing an estimator, name the assumptions under which a coefficient may be interpreted causally, and report how much unmeasured confounding would overturn the conclusion. Drawing and sharing directed acyclic graphs is a useful addition to the researcher toolkit (Lipsky & Greenland, 2022; Tennant et al., 2021).
2. Collect intensive longitudinal process data inside randomized trials, use n-of-1 and micro-randomized designs where within-person causal claims are wanted and effects are reversible, and use randomized multiple-baseline or sequential designs where they are not.
3. Report τ and prediction intervals alongside pooled effects, and state what quantity they summarize: over randomized within-person contrasts they describe variation in effects, over observational coefficients only variation in associations.
4. Before applying an analysis procedure to real data, evaluate the whole procedure using many replicated simulated datasets that match the properties of the data it will be applied to (Morris et al., 2019), and include a homogeneous null to show that it does not manufacture heterogeneity.
5. Report reliability at the two levels the analysis depends on: for the observations, the reliability of within-person change, not the ICC(2) of person means (Cranford et al., 2006); for the person-level estimates, their reliability, and what fraction of their spread is estimation error.

The idionomic program acknowledges what clinicians want to know: what is expected to work for this person rather than for the average person. Yet Hayes et al. never state that as a formal question, and consequently the proposed methods answer a different one. Moving the analysis from the group-level to the person-level changes which associations are estimated, but it does not turn an association into an effect, and it does not make a short series or noisy estimate precise. Those problems are addressed by design and identification of a causal estimand rather than by level of analysis.

Clinical psychology and the idionomic program need a revolution, but not a new one. They need to join the causal revolution, in which the estimand is stated before the data are analyzed.

7 References

- Alahakoon, D., Halgamuge, S. K., & Srinivasan, B. (2000). Dynamic self-organizing maps with controlled growth for knowledge discovery. *IEEE Transactions on Neural Networks*, 11(3), 601–614. <https://doi.org/10.1109/72.846732>
- Barlow, D. H., & Hayes, S. C. (1979). Alternating treatments design: One strategy for comparing the effects of two treatments in a single subject. *Journal of Applied Behavior Analysis*, 12(2), 199–210. <https://doi.org/10.1901/jaba.1979.12-199>
- Berk, R., Brown, L., Buja, A., Zhang, K., & Zhao, L. (2013). Valid post-selection inference. *The Annals of Statistics*, 41(2), 802–837. <https://doi.org/10.1214/12-AOS1077>

- Bolger, N., Zee, K. S., Rossignac-Milon, M., & Hassin, R. R. (2019). Causal processes in psychology are heterogeneous. *Journal of Experimental Psychology: General*, 148(4), 601–618. <https://doi.org/10.1037/xge0000558>
- Borenstein, M., Higgins, J. P. T., Hedges, L. V., & Rothstein, H. R. (2017). Basics of meta-analysis: I^2 is not an absolute measure of heterogeneity. *Research Synthesis Methods*, 8(1), 5–18. <https://doi.org/10.1002/jrsm.1230>
- Bringmann, L. F., Vissers, N., Wichers, M., Geschwind, N., Kuppens, P., Peeters, F., Borsboom, D., & Tuerlinckx, F. (2013). A network approach to psychopathology: New insights into clinical longitudinal data. *PLOS ONE*, 8(4), e60188. <https://doi.org/10.1371/journal.pone.0060188>
- Cranford, J. A., Shrout, P. E., Iida, M., Rafaeli, E., Yip, T., & Bolger, N. (2006). A procedure for evaluating sensitivity to within-person change: Can mood measures in diary studies detect change reliably? *Personality and Social Psychology Bulletin*, 32(7), 917–929. <https://doi.org/10.1177/0146167206287721>
- DeRubeis, R. J., Cohen, Z. D., Forand, N. R., Fournier, J. C., Gelfand, L. A., & Lorenzo-Luaces, L. (2014). The Personalized Advantage Index: Translating Research on Prediction into Individualized Treatment Recommendations. A Demonstration. *PLOS ONE*, 9(1), e83875. <https://doi.org/10.1371/journal.pone.0083875>
- Dunn, G., & Bentall, R. (2007). Modelling treatment-effect heterogeneity in randomized controlled trials of complex interventions (psychological treatments). *Statistics in Medicine*, 26(26), 4719–4745. <https://doi.org/10.1002/sim.2891>
- Fisher, A. J., Medaglia, J. D., & Jeronimus, B. F. (2018). Lack of group-to-individual generalizability is a threat to human subjects research. *Proceedings of the National Academy of Sciences*, 115(27), E6106–E6115. <https://doi.org/10.1073/pnas.1711978115>
- Gates, K. M., Lane, S. T., Varangis, E., Giovanello, K., & Guiskewicz, K. (2017). Unsupervised classification during time-series model building. *Multivariate Behavioral Research*, 52(2), 129–148. <https://doi.org/10.1080/00273171.2016.1256187>
- Gates, K. M., & Molenaar, P. C. M. (2012). Group search algorithm recovers effective connectivity maps for individuals in homogeneous and heterogeneous samples. *NeuroImage*, 63(1), 310–319. <https://doi.org/10.1016/j.neuroimage.2012.06.026>
- Grosz, M. P., Rohrer, J. M., & Thoemmes, F. (2020). The taboo against explicit causal inference in nonexperimental psychology. *Perspectives on Psychological Science*, 15(5), 1243–1255. <https://doi.org/10.1177/1745691620921521>
- Haaf, J. M., & Rouder, J. N. (2019). Some do and some don't? Accounting for variability of individual difference structures. *Psychonomic Bulletin & Review*, 26(3), 772–789. <https://doi.org/10.3758/s13423-018-1522-x>
- Hayes, S. C., Ciarrochi, J., Sahdra, B. K., Hofmann, S. G., Ong, C. W., & Hernández, C. (2026). One size fits none: A call for an idionomic revolution. *Clinical Psychology Review*, 129, 102800. <https://doi.org/10.1016/j.cpr.2026.102800>
- Hernán, M. A., & Robins, J. M. (2020). *Causal inference: What if*. Chapman & Hall/CRC.
- Holland, P. W. (1986). Statistics and Causal Inference. *Journal of the American Statistical Association*, 81(396), 945–960. <https://doi.org/10.1080/01621459.1986.10478354>
- Imai, K., Keele, L., & Tingley, D. (2010). A general approach to causal mediation analysis. *Psychological Methods*, 15(4), 309–334. <https://doi.org/10.1037/a0020761>

- Imai, K., Keele, L., Tingley, D., & Yamamoto, T. (2011). Unpacking the black box of causality: Learning about causal mechanisms from experimental and observational studies. *American Political Science Review*, 105(4), 765–789. <https://doi.org/10.1017/S0003055411000414>
- Kahan, B. C., & Devane, D. (2026). Estimands: What they are and why we should use them. *Journal of Clinical Epidemiology*, 189. <https://doi.org/10.1016/j.jclinepi.2025.112054>
- Kazdin, A. E. (2007). Mediators and mechanisms of change in psychotherapy research. *Annual Review of Clinical Psychology*, 3, 1–27. <https://doi.org/10.1146/annurev.clinpsy.3.022806.091432>
- Kent, D. M., Paulus, J. K., Klaveren, D. van, D’Agostino, R., Goodman, S., Hayward, R., Ioannidis, J. P. A., Patrick-Lake, B., Morton, S., Pencina, M., Raman, G., Ross, J. S., Selker, H. P., Varadhan, R., Vickers, A., Wong, J. B., & Steyerberg, E. W. (2020). The predictive approaches to treatment effect heterogeneity (PATH) statement. *Annals of Internal Medicine*, 172(1), 35–45. <https://doi.org/10.7326/M18-3667>
- Kent, D. M., Steyerberg, E., & Klaveren, D. van. (2018). Personalized evidence based medicine: Predictive approaches to heterogeneous treatment effects. *BMJ*, 363, k4245. <https://doi.org/10.1136/bmj.k4245>
- Klasnja, P., Hekler, E. B., Shiffman, S., Boruvka, A., Almirall, D., Tewari, A., & Murphy, S. A. (2015). Microrandomized trials: An experimental design for developing just-in-time adaptive interventions. *Health Psychology*, 34(S), 1220–1228. <https://doi.org/10.1037/hea0000305>
- Klintwall, L., Bellander, M., & Cervin, M. (2023). Perceived causal problem networks: Reliability, central problems, and clinical utility for depression. *Assessment*, 30(1), 73–83. <https://doi.org/10.1177/10731911211039281>
- Kratochwill, T. R., & Levin, J. R. (2010). Enhancing the scientific credibility of single-case intervention research: Randomization to the rescue. *Psychological Methods*, 15(2), 124–144. <https://doi.org/10.1037/a0017736>
- Kursa, M. B., & Rudnicki, W. R. (2010). Feature selection with the Boruta package. *Journal of Statistical Software*, 36(11), 1–13. <https://doi.org/10.18637/jss.v036.i11>
- LeBreton, J. M., & Senter, J. L. (2008). Answers to 20 questions about interrater reliability and interrater agreement. *Organizational Research Methods*, 11(4), 815–852. <https://doi.org/10.1177/1094428106296642>
- Lemmens, L. H. J. M., Müller, V. N. L. S., Arntz, A., & Huibers, M. J. H. (2016). Mechanisms of change in psychotherapy for depression: An empirical update and evaluation of research aimed at identifying psychological mediators. *Clinical Psychology Review*, 50, 95–107. <https://doi.org/10.1016/j.cpr.2016.09.004>
- Lillie, E. O., Patay, B., Diamant, J., Issell, B., Topol, E. J., & Schork, N. J. (2011). The n-of-1 clinical trial: The ultimate strategy for individualizing medicine? *Personalized Medicine*, 8(2), 161–173. <https://doi.org/10.2217/pme.11.7>
- Lipsky, A. M., & Greenland, S. (2022). Causal Directed Acyclic Graphs. *JAMA*, 327(11), 1083–1084. <https://doi.org/10.1001/jama.2022.1816>
- Lundberg, I., Johnson, R., & Stewart, B. M. (2021). What is your estimand? Defining the target quantity connects statistical evidence to theory. *American Sociological Review*, 86(3), 532–565. <https://doi.org/10.1177/00031224211004187>
- Molenaar, P. C. M. (2004). A manifesto on psychology as idiographic science: Bringing the person back into scientific psychology, this time forever. *Measurement: Interdisciplinary Research and Perspectives*, 2(4), 201–218. https://doi.org/10.1207/s15366359mea0204_1

- Morris, T. P., White, I. R., & Crowther, M. J. (2019). Using simulation studies to evaluate statistical methods. *Statistics in Medicine*, 38(11), 2074–2102. <https://doi.org/10.1002/sim.8086>
- Murphy, S. A. (2005). An experimental design for the development of adaptive treatment strategies. *Statistics in Medicine*, 24(10), 1455–1481. <https://doi.org/10.1002/sim.2022>
- Robins, J. M., Hernán, M. Á., & Brumback, B. (2000). Marginal structural models and causal inference in epidemiology. *Epidemiology*, 11(5), 550–560. <https://doi.org/10.1097/00001648-200009000-00011>
- Rohrer, J. M., & Murayama, K. (2023). These are not the effects you are looking for: Causality and the within-/between-persons distinction in longitudinal data analysis. *Advances in Methods and Practices in Psychological Science*, 6(1). <https://doi.org/10.1177/25152459221140842>
- Rohrer, J. M., Seifert, I. S., Arslan, R. C., Sun, J., & Schmukle, S. C. (2024). The effects of satisfaction with different domains of life on general life satisfaction vary between individuals (but we cannot tell you why). *Collabra: Psychology*, 10(1), 121238. <https://doi.org/10.1525/collabra.121238>
- Ryan, O., Bringmann, L. F., & Schuurman, N. K. (2022). The challenge of generating causal hypotheses using network models. *Structural Equation Modeling: A Multidisciplinary Journal*, 29(6), 953–970. <https://doi.org/10.1080/10705511.2022.2056039>
- Sahdra, B. K., Ciarrochi, J., Klimczak, K. S., Krafft, J., Hayes, S. C., & Levin, M. (2024). Testing the applicability of idionomic statistics in longitudinal studies: The example of “doing what matters.” *Journal of Contextual Behavioral Science*, 32, 100728. <https://doi.org/10.1016/j.jcbs.2024.100728>
- Sahdra, B. K., Woolley, M. G., Hernández, C., Li, W., Hayes, S. C., Ciarrochi, J., & Levin, M. E. (2026). Time series machine learning for idionomic process-based treatment planning: A tutorial on tsBoruta. *Journal of Contextual Behavioral Science*, 100983. <https://doi.org/10.1016/j.jcbs.2026.100983>
- Sanford, B. T., Ciarrochi, J., Hofmann, S. G., Chin, F., Gates, K. M., & Hayes, S. C. (2022). Toward empirical process-based case conceptualization: An idionomic network examination of the process-based assessment tool. *Journal of Contextual Behavioral Science*, 25, 10–25. <https://doi.org/10.1016/j.jcbs.2022.05.006>
- Strobl, C., Boulesteix, A.-L., Zeileis, A., & Hothorn, T. (2007). Bias in random forest variable importance measures: Illustrations, sources and a solution. *BMC Bioinformatics*, 8, 25. <https://doi.org/10.1186/1471-2105-8-25>
- Tennant, P. W. G., Murray, E. J., Arnold, K. F., Berrie, L., Fox, M. P., Gadd, S. C., Harrison, W. J., Keeble, C., Ranker, L. R., Textor, J., Tomova, G. D., Gilthorpe, M. S., & Ellison, G. T. H. (2021). Use of directed acyclic graphs (DAGs) to identify confounders in applied health research: Review and recommendations. *International Journal of Epidemiology*, 50(2), 620–632. <https://doi.org/10.1093/ije/dyaa213>
- Vuorre, M., Kay, M., & Bolger, N. (2024). *Communicating causal effect heterogeneity*. OSF. <https://doi.org/10.31234/osf.io/mwg4f>
- Zhao, Q., & Hastie, T. (2021). Causal interpretations of black-box models. *Journal of Business & Economic Statistics*, 39(1), 272–281. <https://doi.org/10.1080/07350015.2019.1624293>

8 Declaration of generative AI and AI-assisted technologies in the writing process

During the preparation of this work some authors used Claude and Qwen to draft, structure and edit text, to check grammar and spelling, and to format the manuscript. After using these tools, the authors reviewed and edited the content as needed and take full responsibility for the content of the published article.